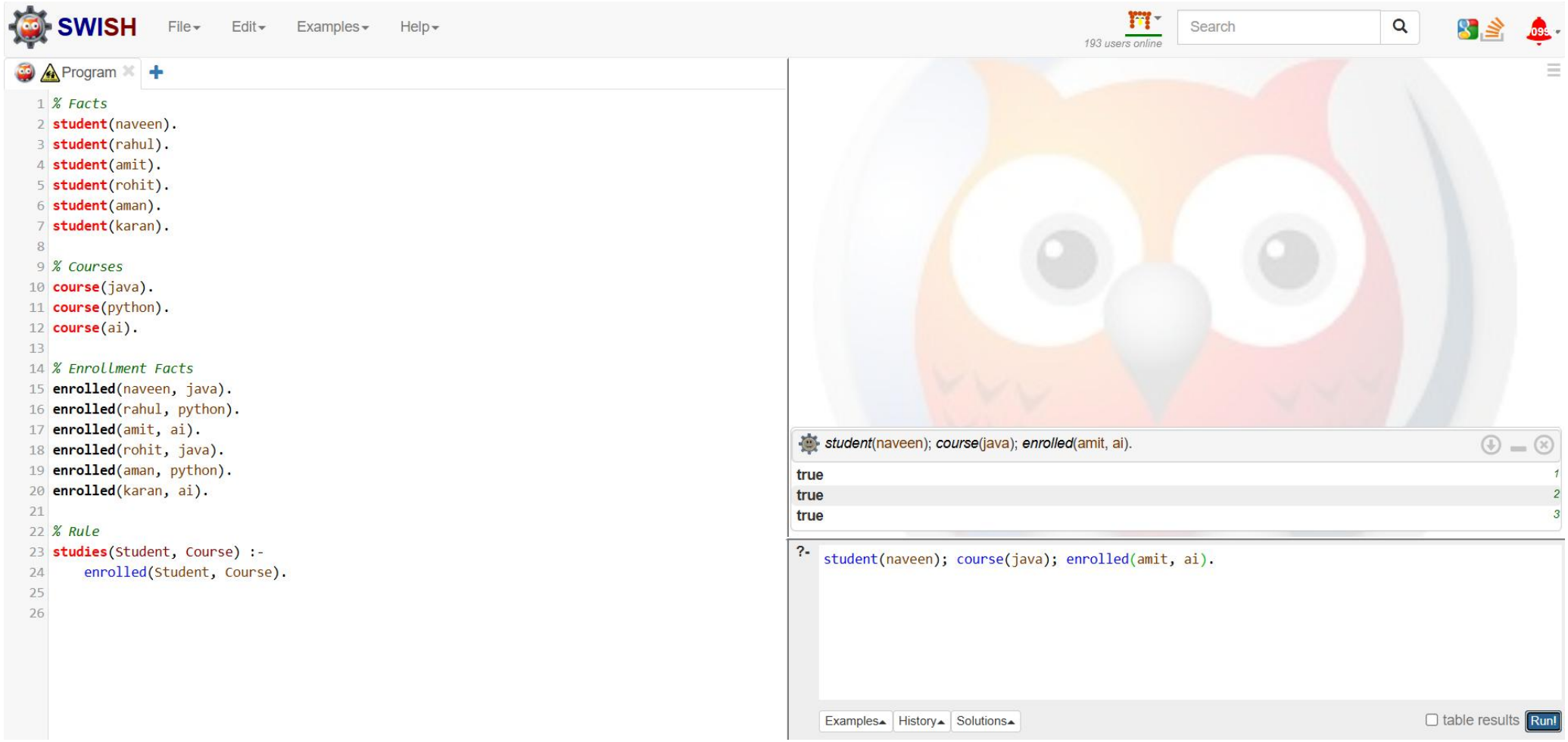
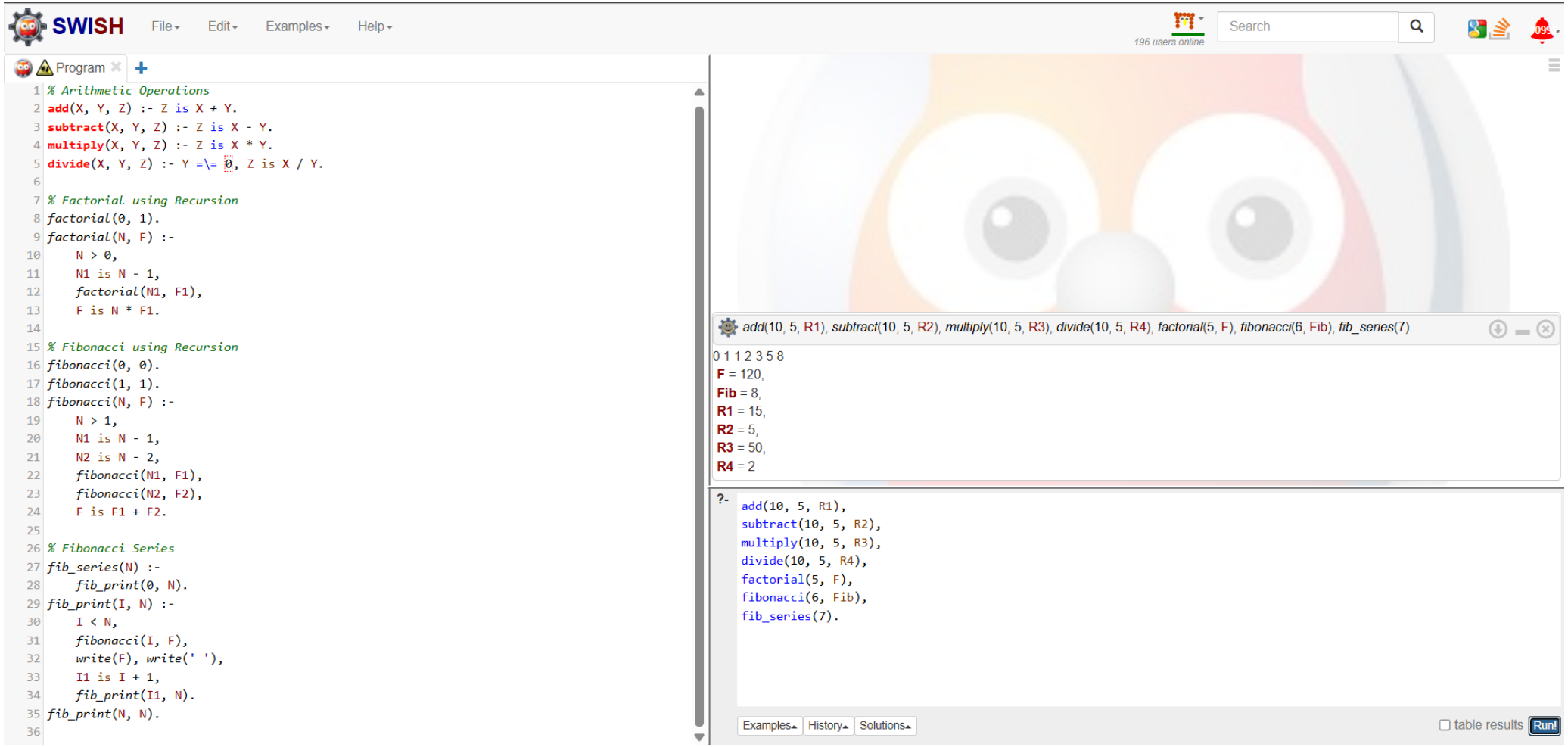
## Experiment - 01

Create a simple Knowledge Base using facts, predicates, clauses, and goals (Student-Course Database).



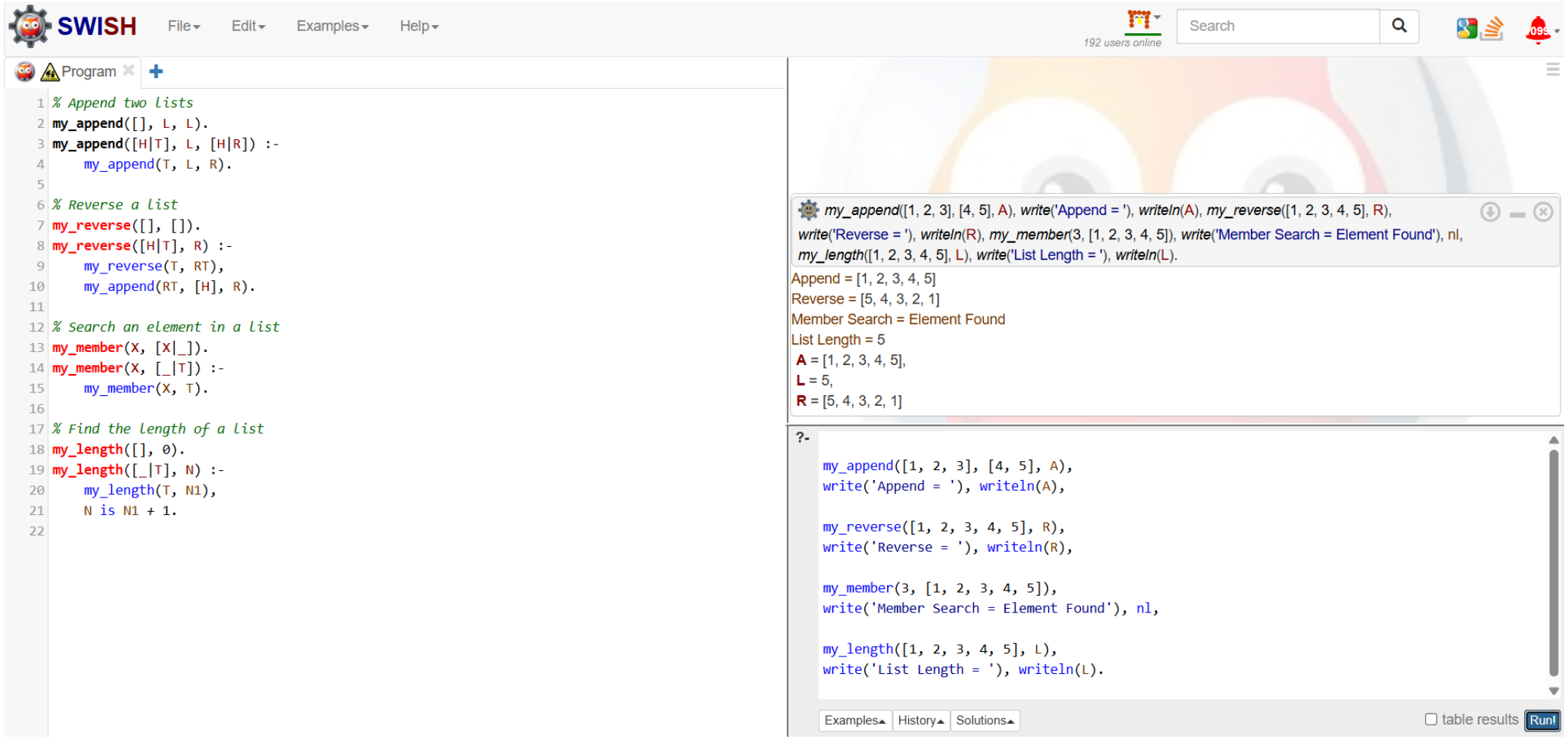
# Experiment – 02

Implement arithmetic operations, Factorial, and Fibonacci Series using recursion.



## Experiment – 03

Perform append, reverse, member search, and list length operations.



## Experiment – 04

Develop a timetable database and answer queries regarding faculty, subjects, and classrooms.

SWISH File Edit Examples Help 192 users online Search

Program

```
% Timetable Database
% timetable(Day, Time, Subject, Faculty, Classroom).

timetable(monday, 9, java, amit, room101).
timetable(monday, 10, dsa, rahul, room102).
timetable(monday, 11, ai, priya, room103).

timetable(tuesday, 9, dsa, rahul, room102).
timetable(tuesday, 10, java, amit, room101).
timetable(tuesday, 11, python, neha, room104).

timetable(wednesday, 9, ai, priya, room103).
timetable(wednesday, 10, python, neha, room104).
timetable(wednesday, 11, dsa, rahul, room102).

% Find faculty teaching a subject
faculty_subject(Faculty, Subject) :-
    timetable(_, _, Subject, Faculty, _).

% Find subjects taught by a faculty
faculty_teaches(Faculty, Subject) :-
    timetable(_, _, Subject, Faculty, _).

% Find classroom for a subject
subject_room(Subject, Room) :-
    timetable(_, _, Subject, _, Room).

% Find timetable for a particular day
day_timetable(Day, Time, Subject, Faculty, Room) :-
    timetable(Day, Time, Subject, Faculty, Room).

```

*faculty_subject*(F1, java), *write*('Faculty teaching Java = '), *writeln*(F1), *faculty_teaches*(rahul, S1), *write*('Subject taught by Rahul = '), *writeln*(S1), *subject_room*(ai, R1), *write*('Classroom for AI = '), *writeln*(R1), *day_timetable*(monday, T, S, F, R), *write*('Monday timetable = '), *write*(T), *write*(' '), *write*(S), *write*(' '), *write*(F), *write*(' '), *writeln*(R).

```
Faculty teaching Java = amit
Subject taught by Rahul = dsa
Classroom for AI = room103
Monday timetable = 9 java amit room101
 F = F1, F1 = amit,
 R = room101,
 R1 = room103,
 S = java,
 S1 = dsa,
 T = 9
```

?-

```
faculty_subject(F1, java),
write('Faculty teaching Java = '), writeln(F1),

faculty_teaches(rahul, S1),
write('Subject taught by Rahul = '), writeln(S1),

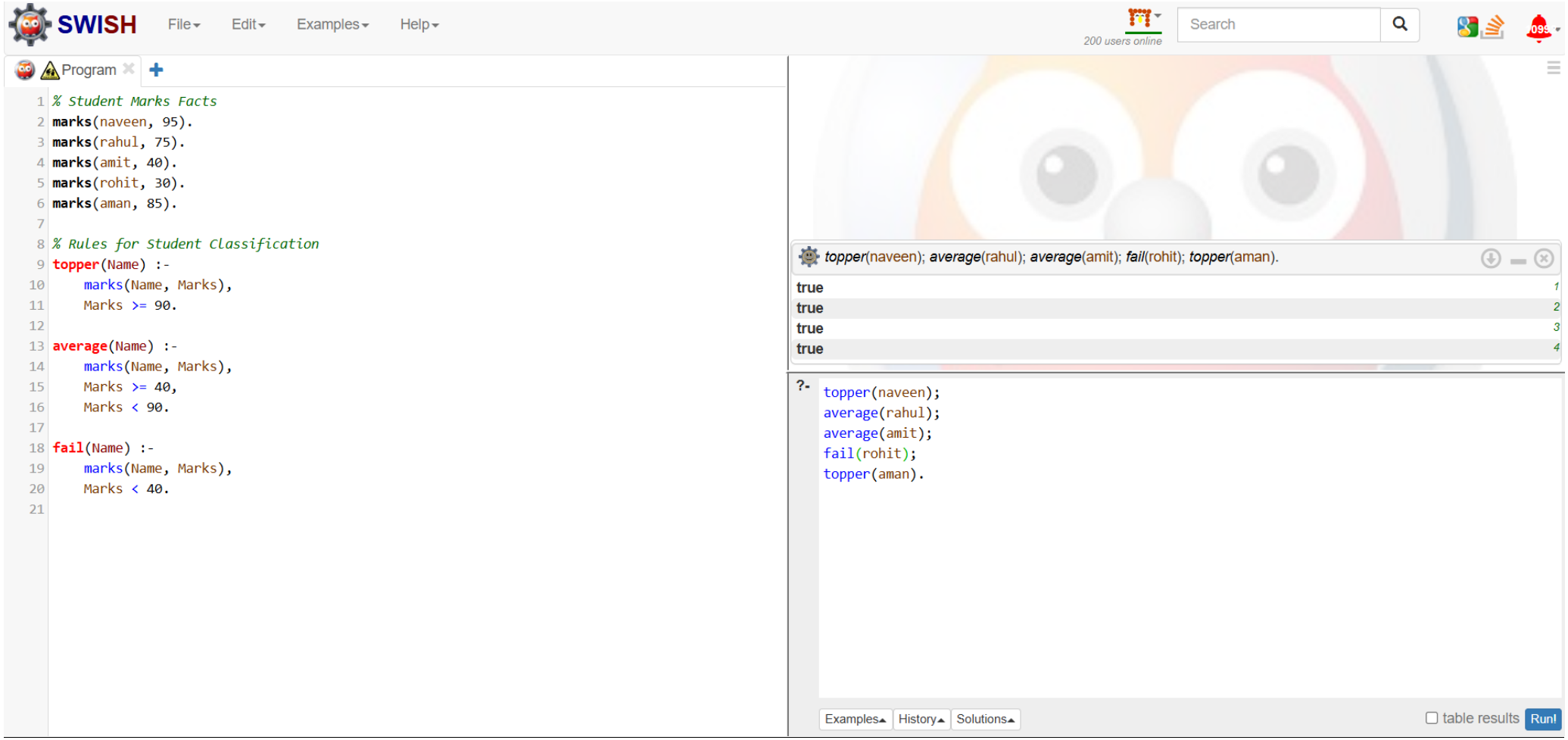
subject_room(ai, R1),
write('Classroom for AI = '), writeln(R1),

day_timetable(monday, T, S, F, R),
write('Monday timetable = '),
write(T), write(' '),
write(S), write(' '),
write(F), write(' '),
writeln(R).
```

Examples History Solutions table results Run!

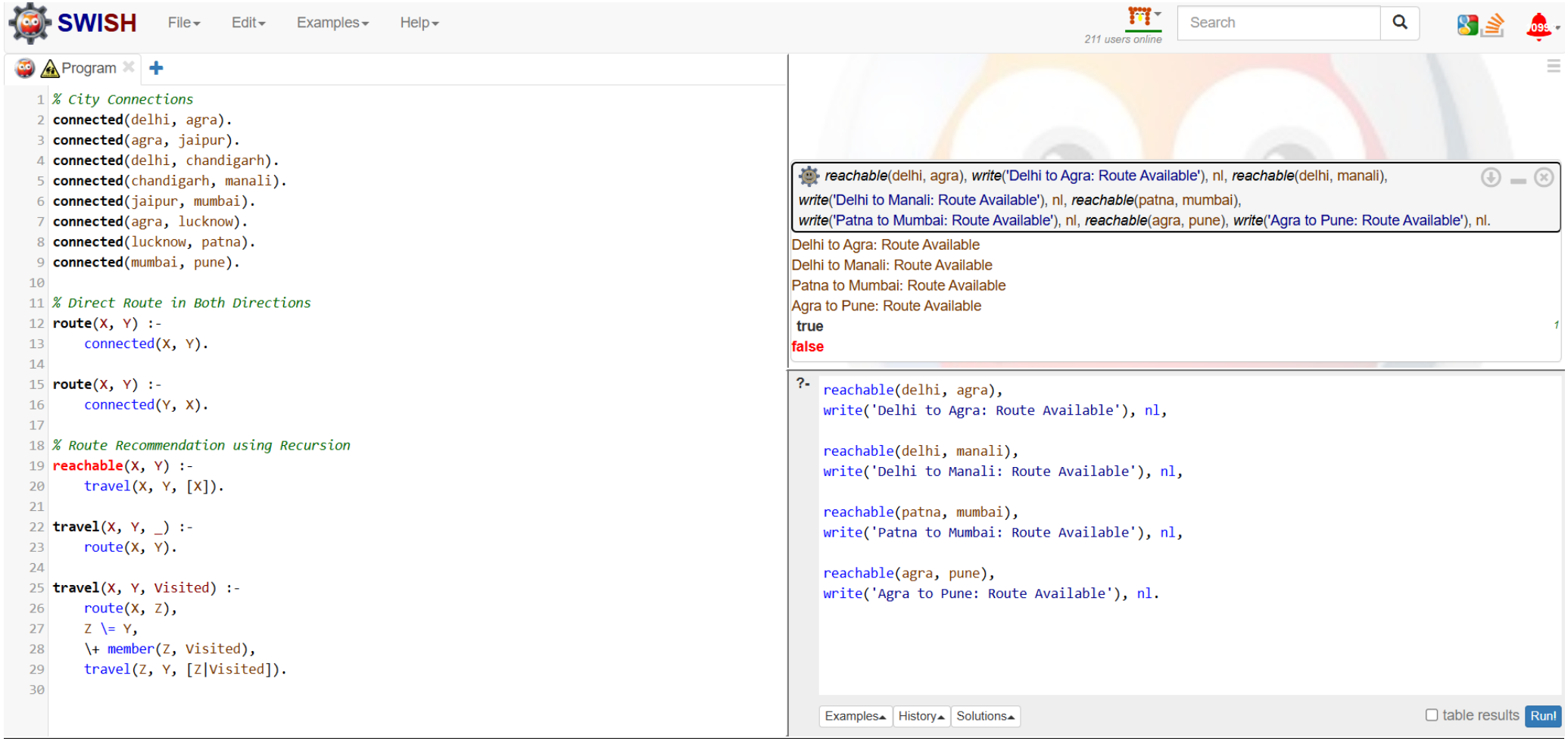
## Experiment – 05

Classify students as topper, average, and fail using rule-based reasoning.



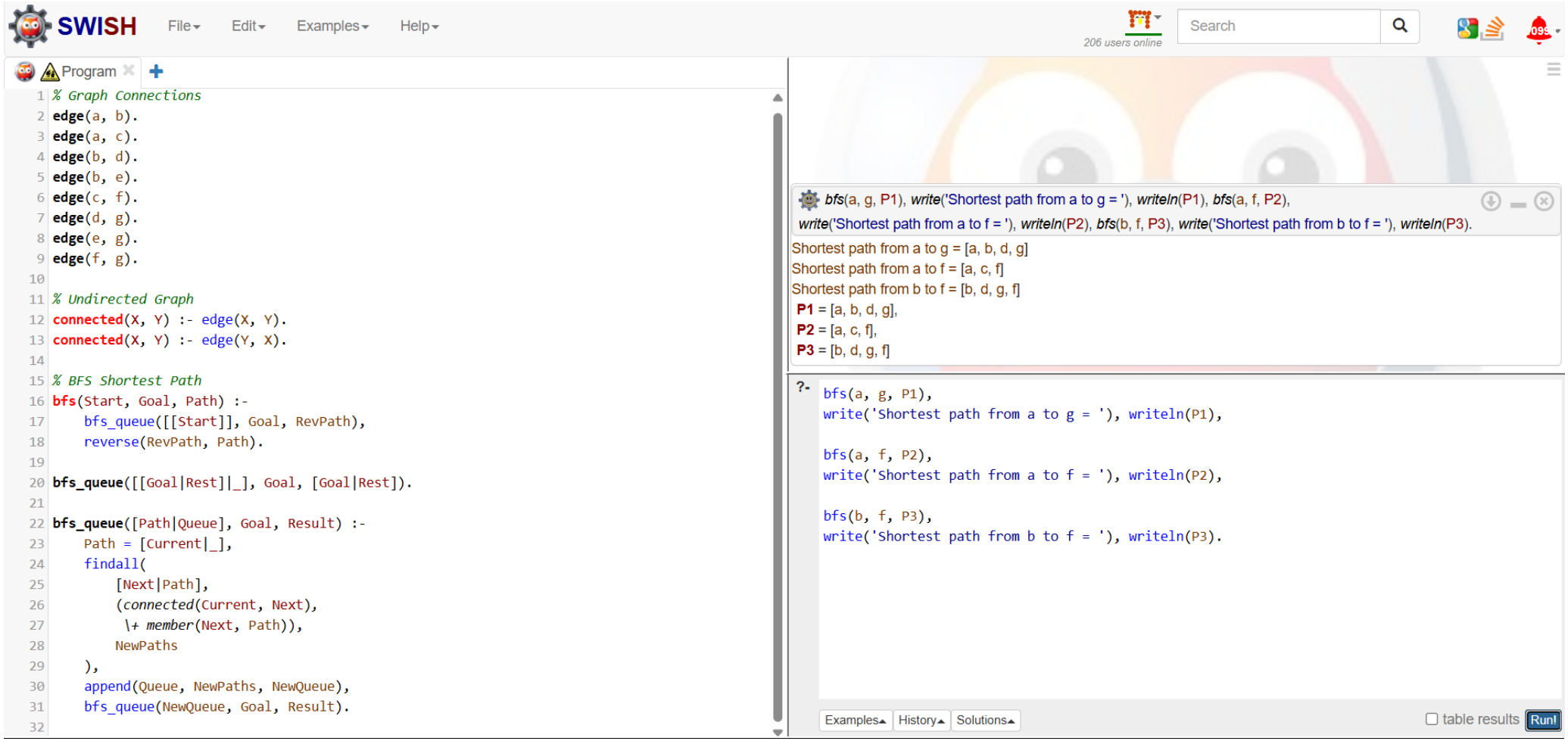
## Experiment – 06

Design a route recommendation system using city connections and inference rules.



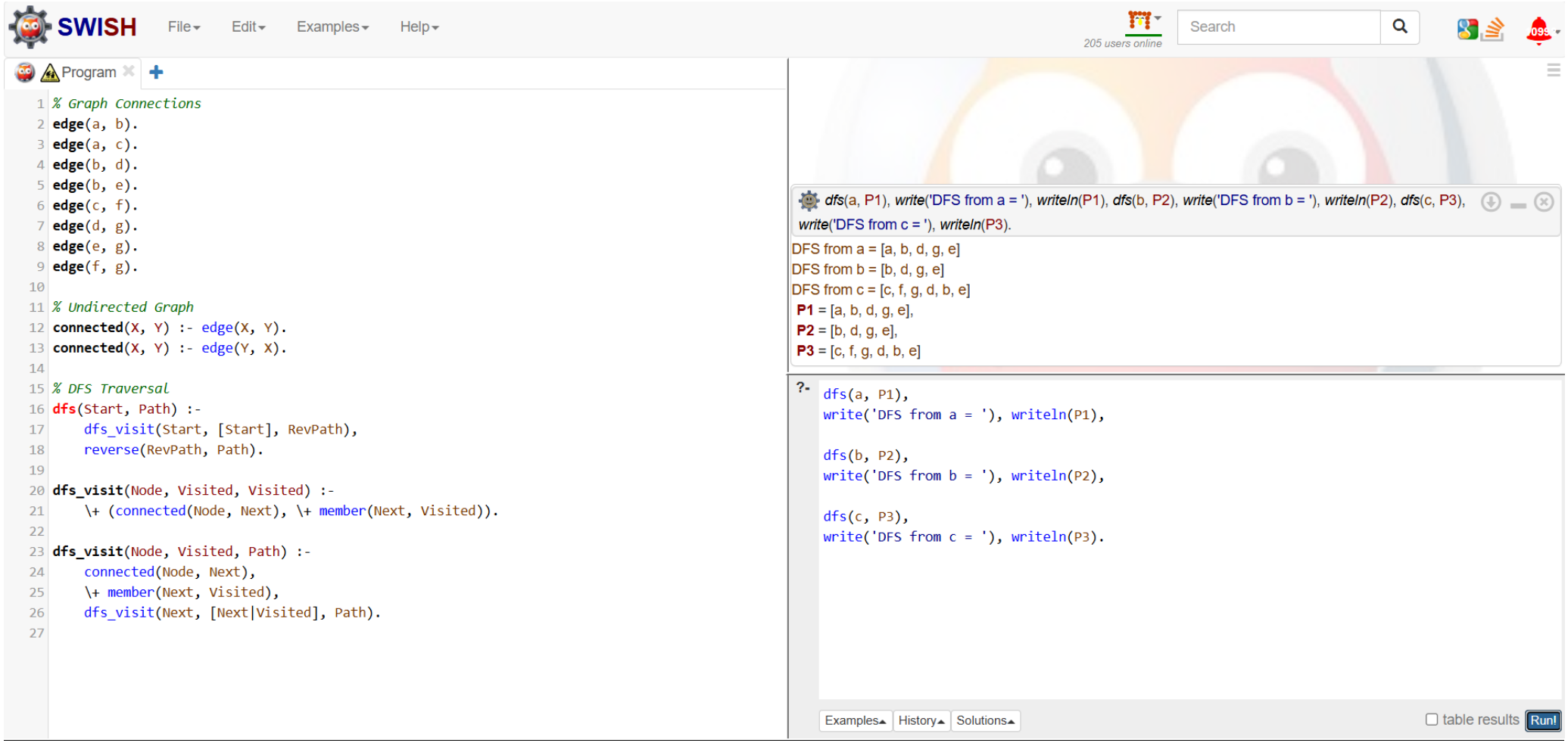
## Experiment – 07

Implement BFS to find shortest paths in a graph.



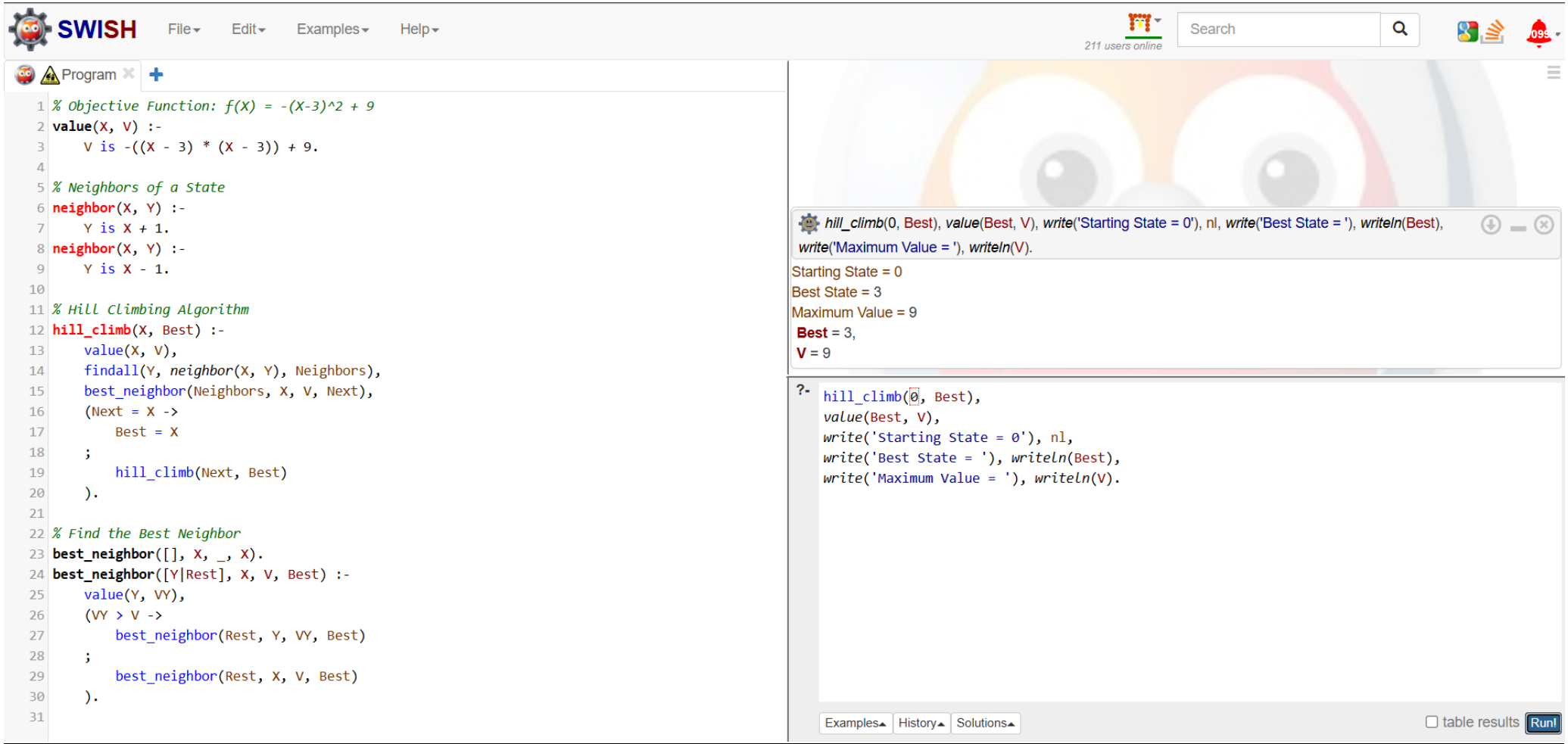
## Experiment – 08

Implement DFS to explore state-space trees and graphs.



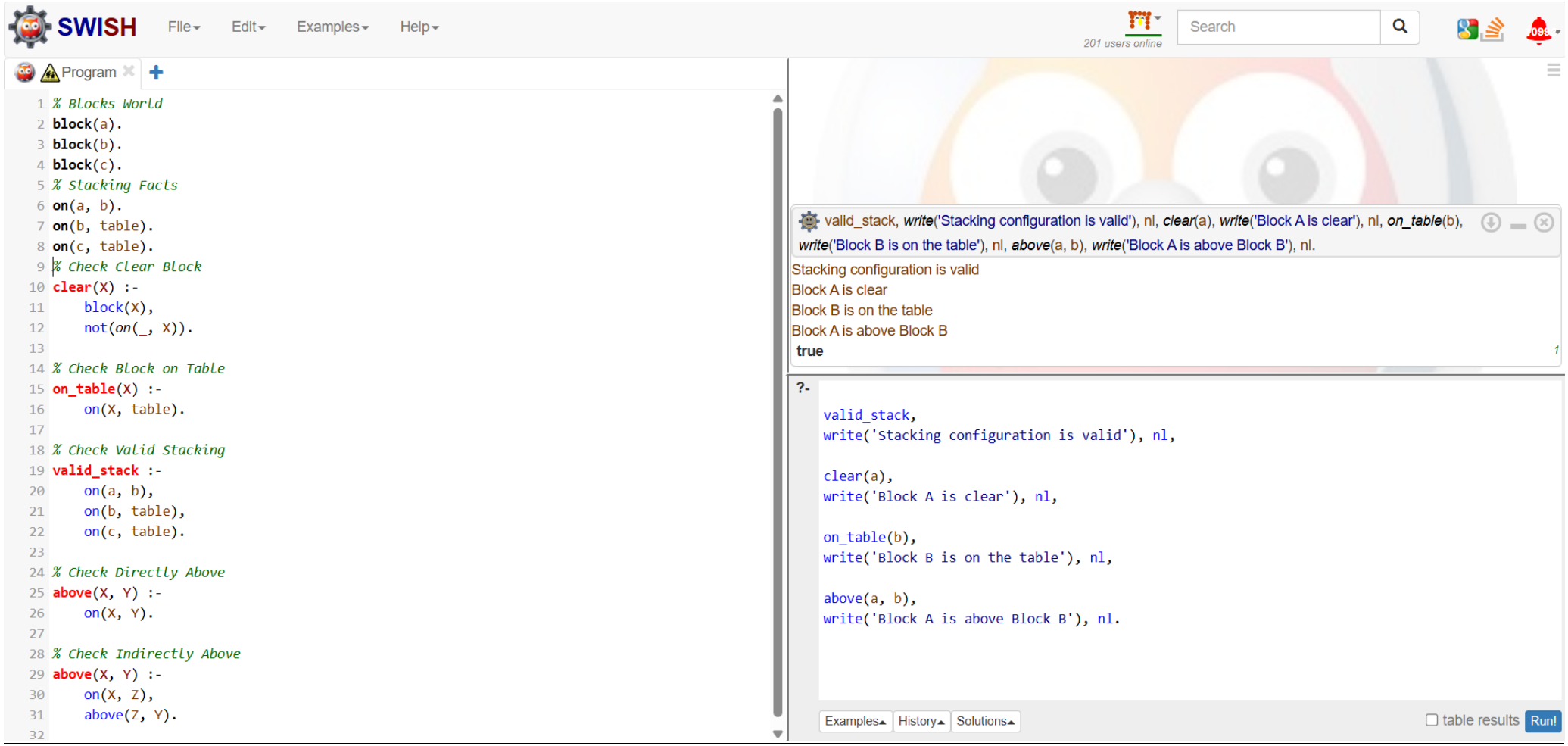
## Experiment – 09

Implement Hill Climbing for optimization problems.



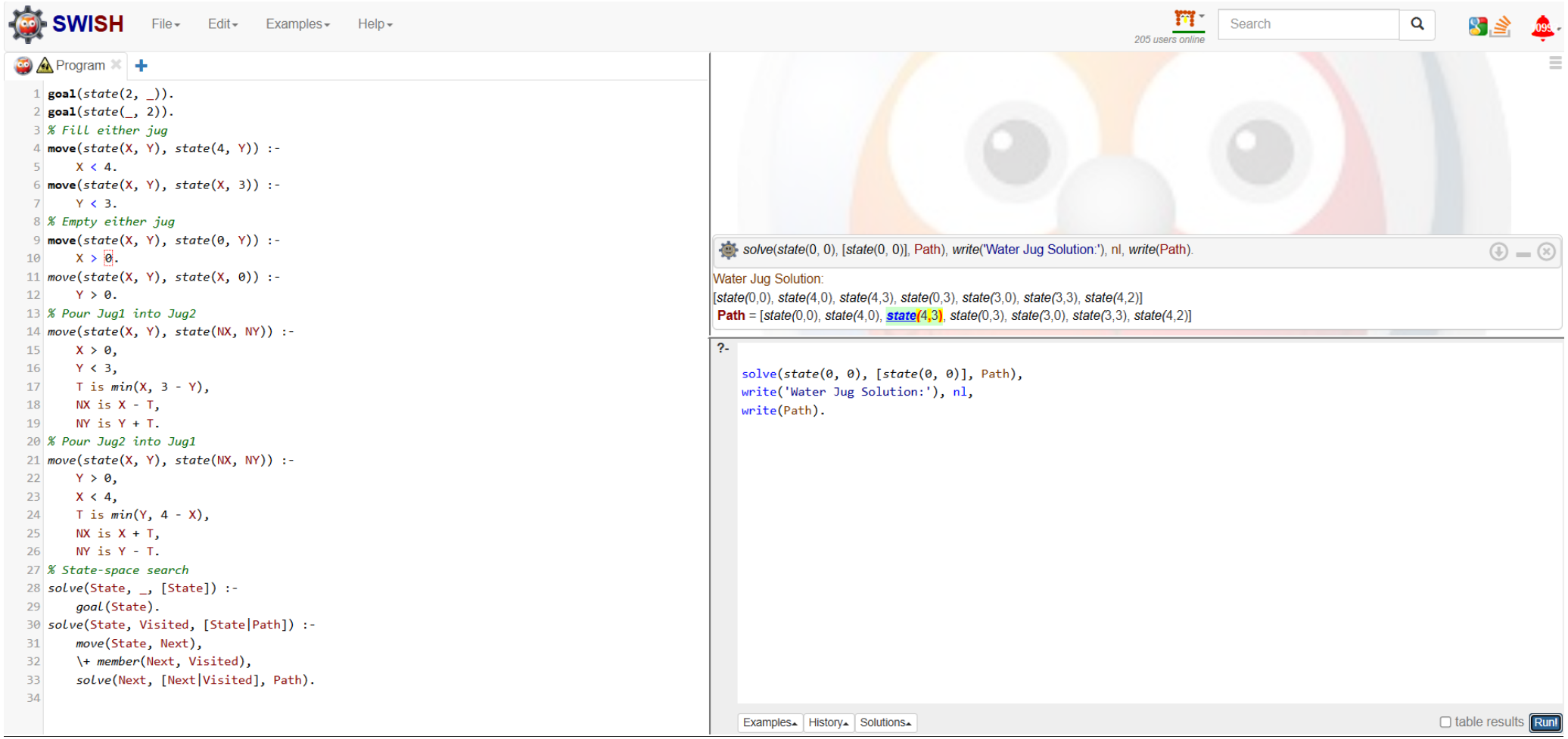
## Experiment – 10

Represent Blocks World and validate stacking configurations using predicates.



## Experiment – 11

Solve the Water Jug Problem using state-space search.



## Experiment – 12

Develop a simple game-playing agent using adversarial search concepts.

